Introduction to Machine Learning

Step 1 – Problem Understanding
↓
Step 2 – Data Collection
↓ EDA
Step 3 – Data Preprocessing
↓
Step 4 – EDA
↓
Step 5 – Modeling
↓
Step 6 – Evaluation
↓
Step 7 – Deployment

Introduction to Machine Learning

Definition of Machine Learning (ML)

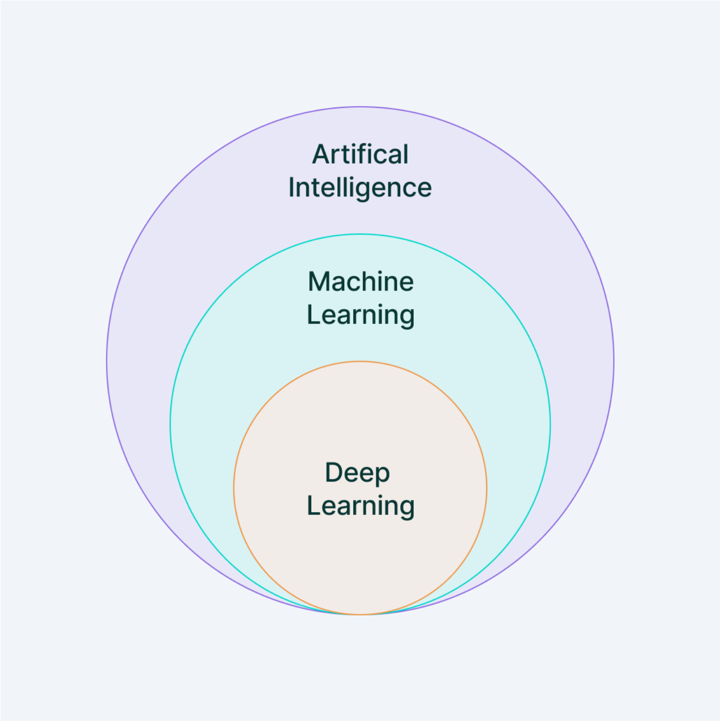
Differences between AI, ML, and Deep Learning

Real-world applications of ML (e.g., image recognition, natural language processing, recommendation systems)

Machine Learning

Machine Learning (ML) is a branch of artificial intelligence (AI) that focuses on building systems that can learn from data, identify patterns, and make decisions with minimal human intervention.

In essence, machine learning allows computers to "learn" from past experiences (data) and improve their performance over time without being explicitly programmed for every possible scenario.



https://www.v7labs.com/blog/machine-learning-guide

Machine Learning: Applications

Image Recognition:

Facial Recognition: Used in smartphones, social media platforms (e.g., Facebook photo tagging), and security systems. ML algorithms can detect and recognize human faces in images or videos.

Medical Imaging: ML is used in analyzing X-rays, MRIs, and CT scans to identify abnormalities such as tumors, fractures, or other diseases.

Object Detection: Used in autonomous vehicles, retail (e.g., Amazon Go stores), and security surveillance systems.

Machine Learning: Applications

Speech and Voice Recognition:

Virtual Assistants: ML powers virtual assistants like Siri, Alexa, and Google Assistant, enabling them to recognize and respond to voice commands.

Speech-to-Text: Applications like Google Voice Typing and automated transcription services convert spoken words into written text.

Voice Biometrics: Used in security systems for identifying individuals based on their voice.

Machine Learning: Applications

Natural Language Processing (NLP):

Chatbots and Customer Support: ML-driven chatbots can understand and respond to customer queries, improving customer service for companies like banks, e-commerce platforms, and service providers.

Language Translation: Google Translate and other language translation apps use ML models to translate text and speech between different languages.

Sentiment Analysis: Businesses use ML to analyze customer reviews, social media posts, and other text data to determine public sentiment or opinions about products or services.

Machine Learning: Applications

Recommendation Systems:

E-commerce: Platforms like Amazon and Alibaba use ML to recommend products based on a user's browsing history, purchase behavior, and preferences.

Streaming Services: Netflix, Spotify, and YouTube recommend movies, music, and videos by analyzing user interactions, viewing history, and preferences.

Social Media: Facebook, Twitter, and Instagram use recommendation algorithms to suggest friends, posts, or content that align with a user's interests.

Machine Learning: Applications

Fraud Detection:

Banking and Finance: ML algorithms analyze transaction patterns to detect fraudulent activities, such as credit card fraud, money laundering, or account takeovers.

Insurance: Insurers use ML to detect fraudulent insurance claims by spotting unusual patterns or behaviors in claim submissions.

Autonomous Vehicles:

Self-Driving Cars: Companies like Tesla, Waymo, and Uber use ML to help autonomous vehicles interpret their surroundings (e.g., detecting pedestrians, road signs, and other cars), make driving decisions, and navigate safely.

Drone Navigation: ML is used in drones for route planning, object detection, and collision avoidance.

Machine Learning: Applications

Healthcare and Medical Diagnosis:

Disease Prediction: ML models help predict the likelihood of diseases such as diabetes, heart disease, or cancer by analyzing patient data.

Personalized Medicine: Machine learning enables the development of personalized treatment plans based on a patient's genetics, medical history, and current health data.

Drug Discovery: ML accelerates the process of discovering new drugs by analyzing data from clinical trials and identifying potential drug candidates.

More….

- Finance and Stock Market Analysis
- Marketing and Advertising
- Supply Chain Optimization
- Gaming, Robotics, Cybersecurity
- Personalization
- Energy Efficiency

Types of Machine Learning

Supervised Learning

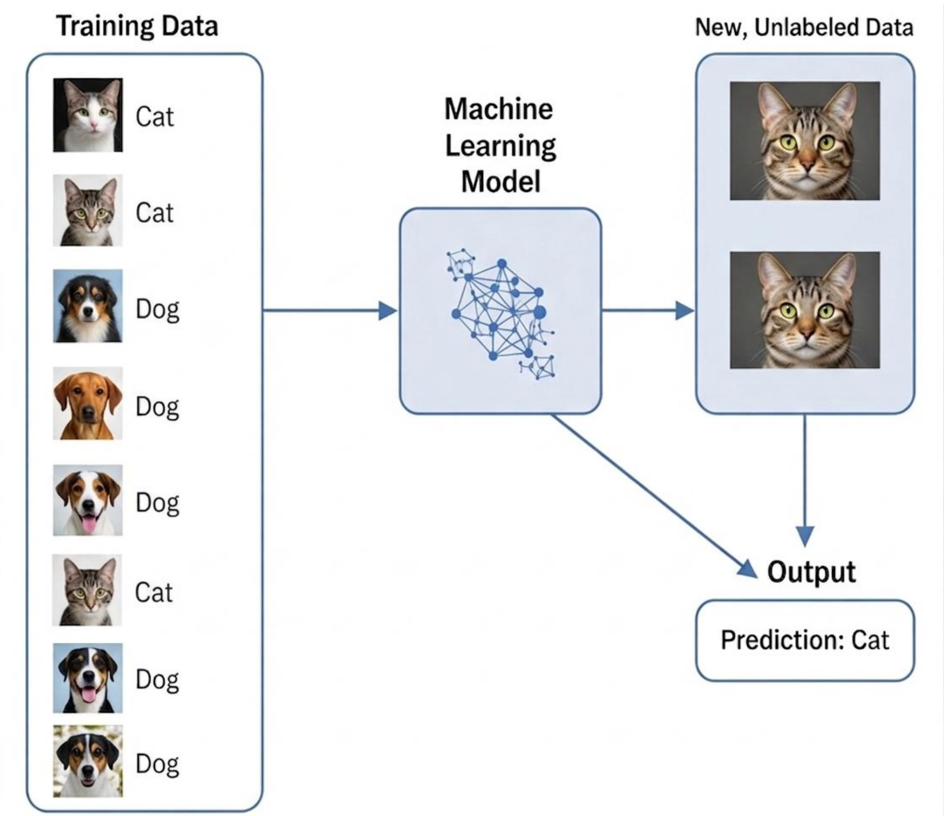
Unsupervised Learning

Types of Machine Learning

Supervised Learning

In supervised learning, the model is trained on **labeled data**, meaning that each training example has both **input data** and a corresponding correct output (**label**).

The goal is for the model to learn the mapping from inputs to outputs so that it can predict the output for new, unseen data.



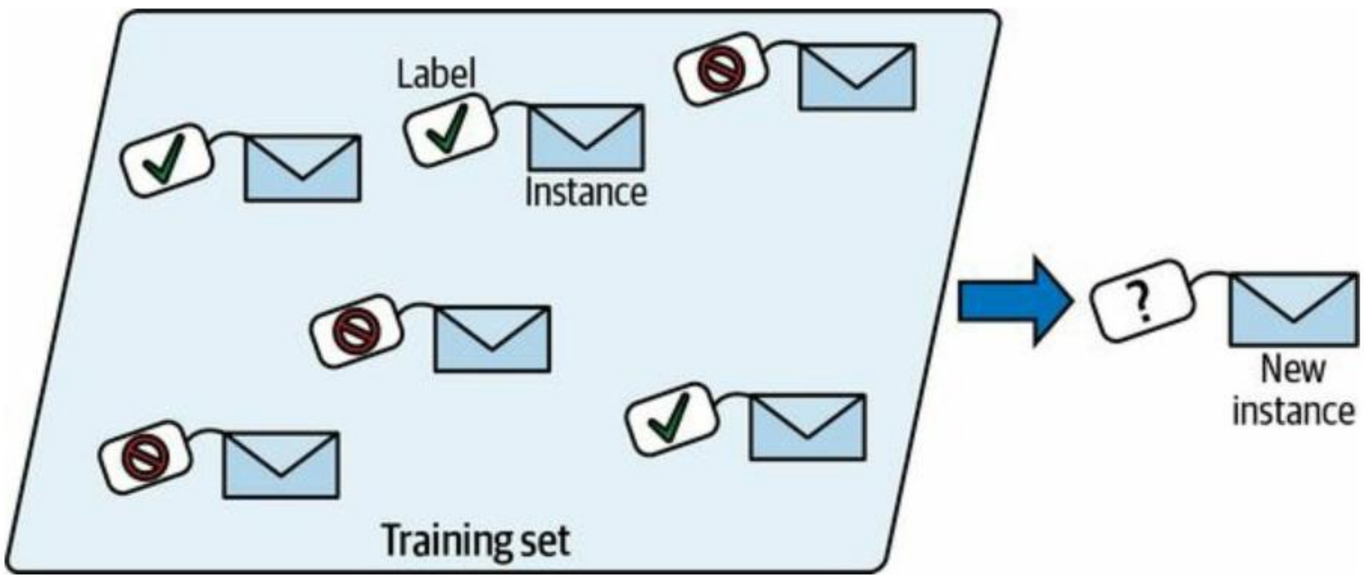
Types of Machine Learning

Supervised Learning

In supervised learning, the model is trained on **labeled data**, meaning that each training example has both **input data** and a corresponding correct output (**label**).

The goal is for the model to learn the mapping from inputs to outputs so that it can predict the output for new, unseen data.

Example: Spam detection in emails, where the model is trained on labeled emails (spam or not spam).



Types of Machine Learning

Supervised Learning: Classification

User ID	Gender	Age	Salary	Purchased
15624510	Male	19	19000	0
15810944	Male	35	20000	1
15668575	Female	26	43000	0
15603246	Female	27	57000	0
15804002	Male	19	76000	1
15728773	Male	27	58000	1
15598044	Female	27	84000	0
15694829	Female	32	150000	1
15600575	Male	25	33000	1
15727311	Female	35	65000	0
15570769	Female	26	80000	1
15606274	Female	26	52000	0
15746139	Male	20	86000	1
15704987	Male	32	18000	0
15628972	Male	18	82000	0
15697686	Male	29	80000	0
15733883	Male	47	25000	1

https://www.geeksforgeeks.org/machine-learning/supervised-machine-learning/

Types of Machine Learning

Supervised Learning: Regression

Temperature	Pressure	Relative Humidity	Wind Direction	Wind Speed
10.69261758	986.882019	54.19337313	195.7150879	3.278597116
13.59184184	987.8729248	48.0648859	189.2951202	2.909167767
17.70494885	988.1119385	39.11965597	192.9273834	2.973036289
20.95430404	987.8500366	30.66273218	202.0752869	2.965289593
22.9278274	987.2833862	26.06723423	210.6589203	2.798230886
24.04233986	986.2907104	23.46918024	221.1188507	2.627005816
24.41475295	985.2338867	22.25082295	233.7911987	2.448749781
23.93361956	984.8914795	22.35178837	244.3504333	2.454271793
22.68800023	984.8461304	23.7538641	253.0864716	2.418341875
20.56425726	984.8380737	27.07867944	264.5071106	2.318677425
17.76400389	985.4262085	33.54900114	280.7827454	2.343950987
11.25680746	988.9386597	53.74139903	68.15406036	1.650191426
14.37810685	989.6819458	40.70884681	72.62069702	1.553469896
18.45114201	990.2960205	30.85038484	71.70604706	1.005017161
22.54895853	989.9562988	22.81738811	44.66042709	0.264133632
24.23155922	988.796875	19.74790765	318.3214111	0.329656571

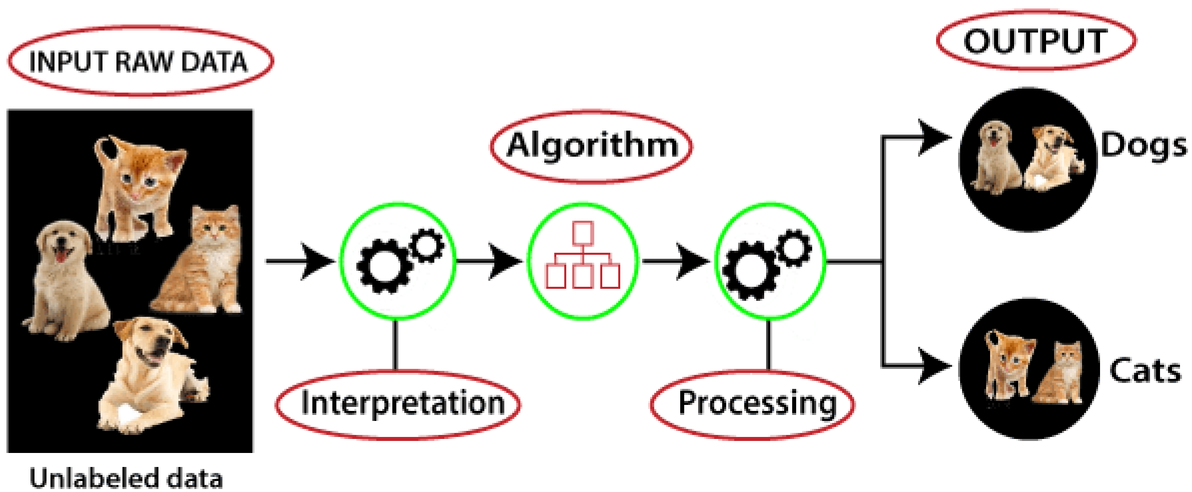
https://www.geeksforgeeks.org/machine-learning/supervised-machine-learning/

Types of Machine Learning

Unsupervised Learning

In unsupervised learning, the model is given data **without explicit labels**. It tries to learn the structure of the data and discover **hidden patterns** or groupings.

Example: Customer segmentation, where the goal is to group customers based on purchasing behavior without knowing any predefined categories.

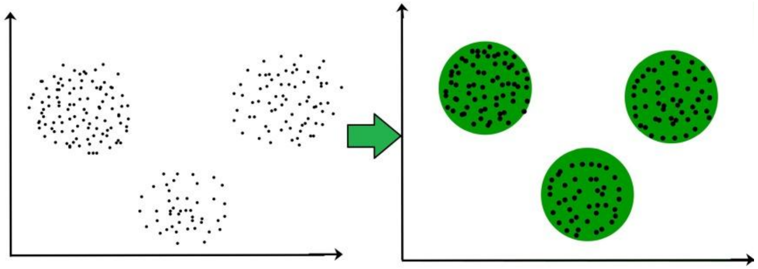


https://www.javatpoint.com/unsupervised-machine-learning

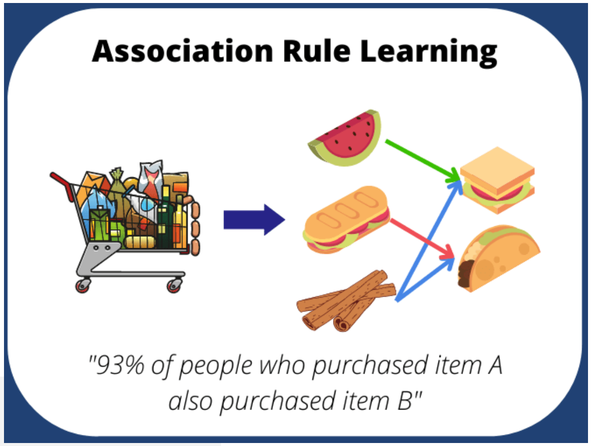
Types of Machine Learning

Unsupervised Learning

- Works with unlabeled data
- Discovers hidden patterns
- Used for exploration and pattern detection
- No supervision (no correct answers provided)



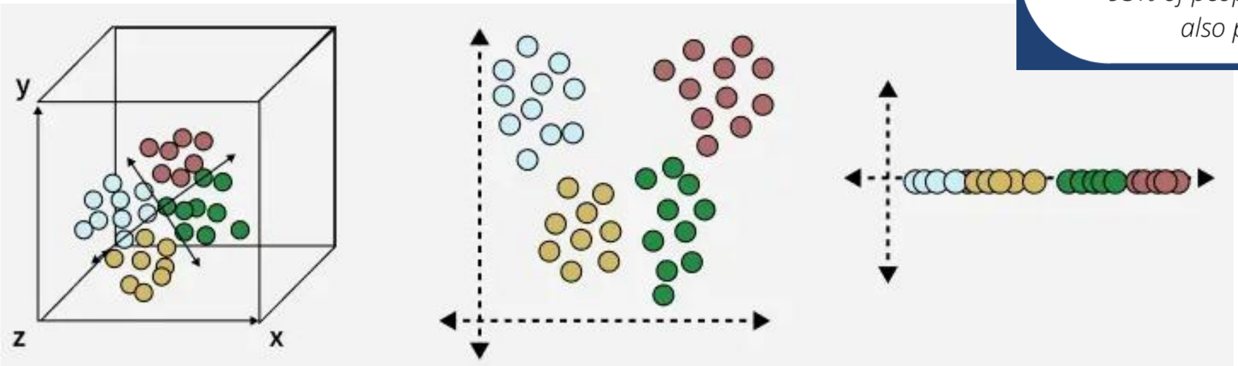
https://www.geeksforgeeks.org/machine-learning/clustering-in-machine-learning/



https://medium.com/@sahinn.muhammed.alii/association-rule-learning-with-apriori-algorithm-649ba26b93b7

Main Types of Unsupervised Learning

- Clustering
- Association Rule Learning
- Dimensionality Reduction



https://www.geeksforgeeks.org/data-science/dimensionality-reduction-techniques/

Key Concepts in Machine Learning

Data and Features: input data, features, labels

Model and Algorithm: what is a model, common ML algorithms

Training and Testing: data splitting (train-test-validation sets)

Loss Function: error measurement, cost function

Optimization: gradient descent basics

Learning Process

How a model learns: data → algorithm → model

Model evaluation metrics: accuracy, precision, recall, F1 score

Overfitting and underfitting: bias-variance tradeoff

Fundamental Algorithms Overview

Linear Regression

Logistic Regression

Decision Trees

k-Nearest Neighbors (k-NN)

Briefly introduce more advanced methods (e.g., Neural Networks, Support Vector Machines)

Challenges in Machine Learning

Data quality and quantity

Feature engineering and selection

Interpretability and explainability of models

Scalability and performance